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Needle-tract assistant including components and methods thereof

Abstract

A needle-tract assistants (**100, 200**) for establishing a needle tract of a predetermined length, including a needle thruster (**110, 210**) and a needle-in-catheter assembly (**130, 230**) removably loaded in the needle thruster (**110, 210**). The needle thruster (**110, 210**) can include a cradle (**112, 212**), a carriage(**114, 214**) within the cradle (**112, 212**), and a plunger (**116, 216**) coupled to the carriages (**114, 214**). The carriages (**114, 214**) can be configured to move between a proximal-end portion and a distal-end portion of the cradle(**112, 212**). The plungers (**116, 216**) can be configured to move the carriages (**114, 214**) within the cradle(**112, 212**). The plungers (**116, 216**) can also be configured to allow a user to set the predetermined length of the needle tract before establishing the needle tract. The needle-in-catheter assembly (**130, 230**) can include a hub (**132, 232**), a catheter tube (**134, 234**) extending from a distal-end portion of the hub (**132, 232**), and a needle (**137, 237**) disposed within the catheter tube (**134, 234**). Also described are components of the needle-tract assistants (**100, 200**) and methods of the needle-tract assistants (**100, 200**) or the components thereof.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4808157	12/1988	Coombs	N/A	N/A
4958901	12/1989	Coombs	N/A	N/A
5360409	12/1993	Boyd, III et al.	N/A	N/A
5441483	12/1994	Avitall	N/A	N/A
5848986	12/1997	Lundquist et al.	N/A	N/A
6117112	12/1999	Mahurkar	604/110	A61M 5/322
6171303	12/2000	Ben-Haim et al.	N/A	N/A
8287481	12/2011	Kahn et al.	N/A	N/A
9901675	12/2017	Ella et al.	N/A	N/A
2002/0082584	12/2001	Rosenman et al.	N/A	N/A
2002/0173751	12/2001	Mastorakis	604/218	A61M 5/34
2006/0259061	12/2005	Kick et al.	N/A	N/A
2007/0191773	12/2006	Wojcik	N/A	N/A
2007/0239119	12/2006	Lipov	N/A	N/A
2009/0177041	12/2008	Stefanchik et al.	N/A	N/A
2011/0137125	12/2010	Belsley	N/A	N/A
2011/0152741	12/2010	Banchieri et al.	N/A	N/A
2011/0202067	12/2010	Falkner et al.	N/A	N/A
2011/0213316	12/2010	Ibrahim et al.	N/A	N/A
2012/0221007	12/2011	Batten et al.	N/A	N/A
2013/0245533	12/2012	Kahn et al.	N/A	N/A
2014/0114330	12/2013	Karasic et al.	N/A	N/A
2014/0171826	12/2013	Lampropoulos et al.	N/A	N/A
2015/0045769	12/2014	Cabrera Aquino	604/117	A61D 7/00
2015/0297246	12/2014	Patel et al.	N/A	N/A

2016/0106971	12/2015	Servin de la Mora Godinez et al.	N/A	N/A
2016/0331443	12/2015	Phan et al.	N/A	N/A
2016/0361088	12/2015	Maguire et al.	N/A	N/A
2017/0049511	12/2016	Uhm et al.	N/A	N/A
2018/0207406	12/2017	Ishida	N/A	N/A
2018/0256021	12/2017	Gill	N/A	N/A
2018/0256223	12/2017	Lueth et al.	N/A	N/A
2018/0344985	12/2017	Shah et al.	N/A	N/A
2019/0117937	12/2018	Humphrey et al.	N/A	N/A
2019/0184136	12/2018	Lubinski et al.	N/A	N/A
2019/0282217	12/2018	Graham et al.	N/A	N/A
2019/0282218	12/2018	Panzenbeck et al.	N/A	N/A
2022/0240981	12/2021	Yi et al.	N/A	N/A
2022/0241562	12/2021	Yi et al.	N/A	N/A
2024/0225692	12/2023	Forreiter	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
107073723	12/2016	CN	N/A
107374702	12/2016	CN	N/A
107412935	12/2016	CN	N/A
107661552	12/2017	CN	N/A
108618739	12/2017	CN	N/A
9724983	12/1996	WO	N/A
98/56293	12/1997	WO	N/A
2008/014791	12/2007	WO	N/A
2008121888	12/2007	WO	N/A
2010115134	12/2009	WO	N/A
14071161	12/2013	WO	N/A
14100349	12/2013	WO	N/A
2018/022432	12/2017	WO	N/A
2020237420	12/2019	WO	N/A
2022242874	12/2021	WO	N/A

OTHER PUBLICATIONS

Bilbao, J., Elorz, M., Vivas, I et al. Transjugular Intrahepatic Portosystemic Shunt (TIPS) in the Treatment of Venous Symptomatic Chronic Portal Thrombosis in Non-cirrhotic Patients. CVIR 27, 474-480 (2004). https://doi.org/10.1007/s00270-004-0241-z (Year: 2004). cited by examiner

PCT/CN2019/088318 filed May 24, 2019 International Search Report and Written Opinion dated Feb. 6, 2020. cited by applicant

EP 19930992.3 filed Nov. 24, 2021 Extended European Search Report dated Oct. 18, 2022. cited by applicant

EP 199310871 filed Nov. 24, 2021 Extended European Search Report dated Nov. 3, 2022. cited by applicant

PCT/CN2019/088334 filed May 24, 2019 International Search Report and Written Opinion dated Feb. 6, 2020. cited by applicant

PCT/CN2019/088347 filed May 24, 2019 International Search Report and Written Opinion dated Feb. 12, 2020. cited by applicant

PCT/EP2021/063619 filed May 21, 2021 International Search Report and Written Opinion dated

Feb. 14, 2022. cited by applicant
U.S. Appl. No. 17/612,188, filed Nov. 17, 2021 Final Office Action dated Oct. 1, 2024. cited by applicant
U.S. Appl. No. 17/612,188, filed Nov. 17, 2021 Non-Final Office Action dated Jun. 20, 2024. cited by applicant
U.S. Appl. No. 17/612,188, filed Nov. 17, 2021 Non-Final Office Action dated Nov. 25, 2024. cited by applicant
U.S. Appl. No. 17/612,196, filed Nov. 17, 2021 Non-Final Office Action dated Nov. 19, 2024. cited by applicant

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Background/Summary

PRIORITY
(1) This application is a U.S. national stage application of International Application No. PCT/CN2019/088318, filed May 24, 2019, which is incorporated by reference in its entirety into this application.
BACKGROUND
(2) In a healthy person, blood flowing from the stomach, esophagus, or intestines first flows through the liver. In an unhealthy person having, for example, liver damage, there can be blood flow-restricting blockages in the liver such that blood cannot easily flow therethrough. Such a condition is known as portal hypertension. Common causes of portal hypertension include alcohol abuse, too much iron in the liver (e.g., hemochromatosis), hepatitis B, hepatitis C, or blood clots in a vein that flows from the liver to the heart. When portal hypertension occurs, the blood flow-restricting blockages can elevate pressure in the portal vein causing it to rupture and seriously bleed. A person with portal hypertension can also have bleeding from the veins of the stomach, esophagus, or intestines (e.g., variceal bleeding), a buildup of fluid in the belly (e.g., ascites), or a buildup of fluid in the chest (e.g., hydrothorax).
(3) Portal hypertension is often treated by way of a percutaneous procedure involving placement of a transjugular intrahepatic portosystemic shunt ("TIPS") between the hepatic vein and the portal vein as shown in FIG. **13** to establish blood flow through the liver. Placement of a portosystemic shunt between the right hepatic vein and the right portal vein is generally preferred. A typical procedure for placing a portosystemic shunt in accordance with the foregoing includes placing an introducer sheath in a distal portion of the right hepatic vein followed by advancing a stiffening cannula, optionally as part of a cannula-in-catheter assembly, through the introducer sheath to the distal portion of the right hepatic vein. A needle-in-catheter assembly is subsequently inserted into the stiffening cannula, the stiffening cannula is wedged against the wall of the right hepatic vein, and the needle-in-catheter assembly is thrust through the liver parenchyma into the right portal vein with a single needle throw. However, such a needle throw is blindly performed. As such, there is a risk of overshooting the portal vein in such procedures, which can introduce complications, prolong the procedures, decrease success rates, and the like.
(4) Disclosed herein is a needle-tract assistant including components and methods thereof that address at least the forgoing shortcomings.
SUMMARY
(5) Disclosed herein is a needle-tract assistant for establishing a needle tract of a predetermined

length. The needle-tract assistant includes, in some embodiments, a needle thruster and a needle-in-catheter assembly removably loaded in the needle thruster. The needle thruster includes a cradle, a carriage within the cradle, and a plunger coupled to the carriage. The carriage is configured to move between a proximal-end portion and a distal-end portion of the cradle. The plunger is configured to move the carriage within the cradle. The plunger is also configured to allow a user to set the predetermined length of the needle tract before establishing the needle tract. The needle-in-catheter assembly includes a hub, a catheter tube extending from a distal-end portion of the hub, and a needle disposed within the catheter tube.

(6) In some embodiments, the cradle of the needle thruster includes a longitudinal opening between the proximal-end portion and the distal-end portion of the cradle. The longitudinal opening is configured to allow the needle-in-catheter assembly to be removed from the needle thruster through the opening.

(7) In some embodiments, the cradle of the needle thruster includes a proximal-end opening at a proximal end of the cradle and a distal-end opening at a distal end of the cradle. The plunger extends through the proximal-end opening of the cradle. The catheter tube and needle extend through the distal-end opening of the cradle when the needle-in-catheter assembly is disposed in the needle thruster.

(8) In some embodiments, the cradle of the needle thruster includes a longitudinal rail extending along an inner surface of the cradle. The longitudinal rail is configured to guide movement of the plunger, the carriage, or both the plunger and the carriage within the cradle.

(9) In some embodiments, the plunger of the needle thruster includes a longitudinal channel extending along an outer surface of the plunger. The longitudinal channel is configured to receive the longitudinal rail of the cradle.

(10) In some embodiments, the cradle of the needle thruster includes a spiral rail spiraling around an inner surface of the cradle. The spiral rail is configured to guide movement of the plunger, the carriage, or both the plunger and the carriage within the cradle. The spiral rail is also configured to govern a linear velocity of the plunger, the carriage, or both the plunger and the carriage within the cradle.

(11) In some embodiments, the cradle of the needle thruster includes a spiral channel spiraling around an outer surface of the plunger. The spiral channel is configured to receive the spiral rail of the cradle.

(12) In some embodiments, the carriage of the needle thruster includes an unloading mechanism for unloading the needle-in-catheter assembly from the carriage.

(13) In some embodiments, the unloading mechanism includes a compression spring-loaded lever shaped to form a proximal-end portion of a receptacle in the carriage. The receptacle is configured to receive the hub of the needle-in-catheter assembly. The lever is configured to compress the spring of the unloading mechanism upon pressing the lever toward the cradle. By pressing the lever toward the cradle, lever assumes a spring force of the unloading-mechanism spring and increases a length of the receptacle, which allows the hub of the needle-in-catheter assembly to be removed from the receptacle.

(14) In some embodiments, the hub of the needle-in-catheter assembly includes a loading mechanism for loading the needle-in-catheter assembly in the carriage.

(15) In some embodiments, the loading mechanism includes a compression spring-loaded proximal-end cap of the hub of the needle-in-catheter assembly. The proximal-end cap is configured to move toward the distal-end portion of the hub and compress the spring of the loading mechanism upon loading the hub in the receptacle of the carriage. A spring force exerted by the loading-mechanism spring holds the hub in the receptacle of the carriage.

(16) In some embodiments, the hub of the needle-in-catheter assembly includes an internal chamber in a proximal-end portion of the hub. A port of the hub and one or more microlumens longitudinally extending through the catheter tube are fluidly connected to the internal chamber.

(17) In some embodiments, the needle of the needle-in-catheter assembly extends through the internal chamber of the hub without a fluid connection with the internal chamber of the hub.
(18) In some embodiments, a distal end of the needle of the needle-in-catheter assembly distally extends beyond a distal end of the catheter tube of the of the needle-in-catheter assembly. A distal-end portion of the catheter tube including the distal end of the catheter tube has a tapered distal tip including one or more openings corresponding to the one or more microlumens of the catheter tube.
(19) Also disclosed herein is a needle-in-catheter assembly for establishing a needle tract. The needle-in-catheter assembly includes a hub, a catheter tube extending from a distal-end portion of the hub, and a needle disposed within the catheter tube. The catheter tube is a double-walled catheter tube including an outer wall and an inner wall. The needle is fixed to a compression spring-loaded proximal-end cap of the hub.
(20) In some embodiments, the needle is disposed within a needle lumen longitudinally extending through the catheter tube. The needle lumen is defined by an inner surface of the inner wall of the catheter tube.
(21) In some embodiments, the catheter tube includes one or more microlumens longitudinally extending through the catheter tube. Each of the one or more microlumens defined by an inner surface of the outer wall of the catheter tube, an outer surface of the inner wall of the catheter tube, and one or more struts longitudinally extending through the catheter tube between the outer wall and the inner wall of the catheter tube.
(22) In some embodiments, the hub includes an internal chamber in a proximal-end portion of the hub. A port of the hub and the one or more microlumens are fluidly connected to the internal chamber.
(23) In some embodiments, the needle extends through the internal chamber of the hub without a fluid connection with the internal chamber.
(24) In some embodiments, a distal end of the needle distally extends beyond a distal end of the catheter tube. A distal-end portion of the catheter tube including the distal end of the catheter tube has a tapered distal tip including one or more openings corresponding to the one or more microlumens of the catheter tube.
(25) In some embodiments, the hub includes a loading mechanism including the proximal-end cap of the hub for loading the needle-in-catheter assembly in a carriage of a needle thruster. The proximal-end cap is configured to move toward the distal-end portion of the hub and compress the spring of the loading mechanism upon loading the hub in a receptacle of the carriage. A spring force exerted by the loading-mechanism spring holds the hub in the receptacle of the carriage.
(26) In some embodiments, the hub includes a longitudinal handle extending from the hub between a proximal-end portion and the distal-end portion of the hub.
(27) Also disclosed herein is a method of establishing a needle tract of a predetermined length with a needle-tract assistant. The method includes, in some embodiments, inserting a needle-in-catheter assembly into a stiffening cannula disposed in an introducer sheath positioned in a hepatic vein, the needle-in-catheter assembly including a needle having a distal end extending beyond that of a catheter surrounding the needle; moving a plunger of a needle thruster to set a carriage coupled to the plunger in position within a cradle of the needle thruster for establishing the needle tract of the predetermined length; loading a hub of the needle-in-catheter assembly in the carriage of the needle thruster; and thrusting the plunger of the needle thruster toward a distal end of the needle thruster to thrust the needle-in-catheter assembly through a liver parenchyma into a portal vein, thereby forming the needle tract of the predetermined length.
(28) In some embodiments, the method further includes radiographically determining a length for the needle tract of the predetermined length before setting the carriage of the needle thruster in position within the cradle for establishing the needle tract of the predetermined length.
(29) In some embodiments, the method further includes radiographically confirming a length of the needle tract after forming the needle tract of the predetermined length by counting a number of

evenly spaced radiopaque rings on the catheter of the needle-in-catheter assembly.
(30) These and other features of the concepts provided herein will become more apparent to those of skill in the art in view of the accompanying drawings and following description, which disclose particular embodiments of such concepts in greater detail.

Description

DRAWINGS
(1) FIG. **1** illustrates a first view of a first needle-tract assistant, in accordance with some embodiments.
(2) FIG. **2** illustrates a second view of the first needle-tract assistant, in accordance with some embodiments.
(3) FIG. **3** illustrates a third view of the first needle-tract assistant, in accordance with some embodiments.
(4) FIG. **4** illustrates a first view of a second needle-tract assistant, in accordance with some embodiments.
(5) FIG. **5** illustrates a second view of the second needle-tract assistant, in accordance with some embodiments.
(6) FIG. **6** illustrates a third view of the second needle-tract assistant, in accordance with some embodiments.
(7) FIG. **7** illustrates a transverse cross section of either the first needle-track assistant or the second needle-tract assistant, in accordance with some embodiments.
(8) FIG. **8** illustrates a first view of a distal-end portion of a needle disposed in a catheter tube, in accordance with some embodiments.
(9) FIG. **9** illustrates a second of a distal-end portion of a needle disposed in a catheter tube, in accordance with some embodiments.
(10) FIG. **10** illustrates a first view of a longitudinal cross section of the first needle-tract assistant, in accordance with some embodiments.
(11) FIG. **11** illustrates a second view of a longitudinal cross section of the first needle-tract assistant, in accordance with some embodiments.
(12) FIG. **12** illustrates a catheter tube-advancing mechanism of a hub of a needle-in-catheter assembly, in accordance with some embodiments.
(13) FIG. **13** illustrates a procedure involving placement of a TIPS between a hepatic vein and a portal vein, in accordance with some embodiments.
DESCRIPTION
(14) Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.
(15) Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to

reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(16) With respect to “proximal,” a “proximal portion” or a “proximal-end portion” of, for example, a catheter disclosed herein includes a portion of the catheter intended to be near a clinician when the catheter is used on a patient. Likewise, a “proximal length” of, for example, the catheter includes a length of the catheter intended to be near the clinician when the catheter is used on the patient. A “proximal end” of, for example, the catheter includes an end of the catheter intended to be near the clinician when the catheter is used on the patient. The proximal portion, the proximal-end portion, or the proximal length of the catheter can include the proximal end of the catheter; however, the proximal portion, the proximal-end portion, or the proximal length of the catheter need not include the proximal end of the catheter. That is, unless context suggests otherwise, the proximal portion, the proximal-end portion, or the proximal length of the catheter is not a terminal portion or terminal length of the catheter.

(17) With respect to “distal,” a “distal portion” or a “distal-end portion” of, for example, a catheter disclosed herein includes a portion of the catheter intended to be near or in a patient when the catheter is used on the patient. Likewise, a “distal length” of, for example, the catheter includes a length of the catheter intended to be near or in the patient when the catheter is used on the patient. A “distal end” of, for example, the catheter includes an end of the catheter intended to be near or in the patient when the catheter is used on the patient. The distal portion, the distal-end portion, or the distal length of the catheter can include the distal end of the catheter; however, the distal portion, the distal-end portion, or the distal length of the catheter need not include the distal end of the catheter. That is, unless context suggests otherwise, the distal portion, the distal-end portion, or the distal length of the catheter is not a terminal portion or terminal length of the catheter.

(18) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art.

(19) As set forth above, portal hypertension is often treated by way of a percutaneous procedure involving placement of a TIPS between the hepatic vein and the portal vein as shown in FIG. **13** to establish blood flow through the liver. Placement of a portosystemic shunt between the right hepatic vein and the right portal vein is generally preferred. A typical procedure for placing a portosystemic shunt in accordance with the foregoing includes placing an introducer sheath in a distal portion of the right hepatic vein followed by advancing a stiffening cannula, optionally as part of a cannula-in-catheter assembly, through the introducer sheath to the distal portion of the right hepatic vein. A needle-in-catheter assembly is subsequently inserted into the stiffening cannula, the stiffening cannula is wedged against the wall of the right hepatic vein, and the needle-in-catheter assembly is thrust through the liver parenchyma into the right portal vein with a single needle throw. However, such a needle throw is blindly performed. As such, there is a risk of overshooting the portal vein in such procedures, which can introduce complications, prolong the procedures, decrease success rates, and the like.

(20) Disclosed herein is a needle-tract assistant including components and methods thereof that address at least the forgoing shortcomings.

(21) FIGS. **1-3** illustrate different views of a first needle-tract assistant **100**, in accordance with some embodiments. FIGS. **4-6** illustrate different views of a second needle-tract assistant **200**, in accordance with some embodiments. FIG. **7** illustrates a transverse cross section of either the first needle-track assistant **100** or the second needle-tract assistant **200**, in accordance with some embodiments. FIGS. **10** and **11** illustrate different views of a longitudinal cross section of the first needle-tract assistant **100**, in accordance with some embodiments. FIG. **12** illustrates a catheter tube-advancing mechanism of a hub **132** of a needle-in-catheter assembly **130** or **230**, in accordance with some embodiments.

(22) The needle-tract assistant **100** or **200** includes a needle thruster **110** or **210** and the needle-in-catheter assembly **130** or **230** removably loaded in the needle thruster **110** or **210**.

(23) Beginning with the needle thruster **110** or **210**, the needle thruster **110** includes a cradle **112** or **212**, a carriage **114** or **214** movably disposed within the cradle **112** or **212**, and a plunger **116** or **216** coupled to the carriage **114** or **214**. Each of the cradle **112** or **212**, the carriage **114** or **214**, and the plunger **116** or **216** will now be described.

(24) The cradle **112** or **212** of the needle thruster **110** or **210** includes a longitudinal opening between a proximal-end portion and a distal-end portion of the cradle **112** or **212**. The longitudinal opening is configured to allow the needle-in-catheter assembly **130** or **230** to be inserted or removed from the needle thruster **110** or **210** through the opening.

(25) The cradle **112** or **212** of the needle thruster **110** or **210** includes a proximal-end opening at a proximal end of the cradle **112** or **212** and a distal-end opening at a distal end of the cradle **112** or **212**. The plunger **116** or **216** extends through the proximal-end opening of the cradle **112** or **212**. A catheter tube **134** or **234** and a needle **137** or **237** extend through the distal-end opening of the cradle **112** or **212** when the needle-in-catheter assembly **130** or **230** is disposed in the needle thruster **110** or **210**. While configured differently for each needle thruster of the needle thrusters **110** and **210**, the distal-end opening at the distal end of the cradle **112** or **212** has smaller dimensions (e.g., diameter, width, etc.) than a distal end of the needle-in-catheter assembly **130** or **230**, thereby providing a stop for the needle-in-catheter assembly **130** or **230**.

(26) The cradle **112** of the needle thruster **110** includes a longitudinal rail **118** extending along an inner surface of the cradle **112**. The longitudinal rail **118** is configured to guide movement of the plunger **116**, the carriage **114**, or both the plunger **116** and the carriage **114** within the cradle **112**. Differently, the cradle **212** of the needle thruster **210** includes a spiral rail **218** spiraling around an inner surface of the cradle **212**. The spiral rail **218** is configured to guide movement of the plunger **216**, the carriage **214**, or both the plunger **216** and the carriage **214** within the cradle **212**. The spiral rail **218** is also configured to govern a linear velocity of the plunger **216**, the carriage **214**, or both the plunger **216** and the carriage **214** within the cradle **212**.

(27) The carriage **114** or **214** of the needle thruster **110** or **210** is configured to move between the proximal-end portion and the distal-end portion of the cradle **112** or **212** such as on a same or different rail than the longitudinal rail **118** or spiral rail **218** along the inner surface of the cradle **112** or **212**.

(28) The carriage **114** or **214** of the needle thruster **110** or **210** includes an unloading mechanism for unloading the needle-in-catheter assembly **130** or **230** from the carriage **114** or **214**. The unloading mechanism includes a compression spring-loaded lever **122** or **222** shaped to form a proximal-end portion of a receptacle **124** or **224** (not shown) in the carriage **114** or **214**. (For the receptacle **224**, see the receptacle **124** of FIGS. **10** and **11**. The receptacle **224** is analogous to the receptacle **124**.) The receptacle **124** or **224** is configured to receive the hub **132** or **232** of the needle-in-catheter assembly **130** or **230**. The lever **122** or **222** is configured to compress the spring **126** or **226** (not shown) of the unloading mechanism upon pressing the lever **122** or **222** toward the cradle **112** or **212**. (For the spring **226**, see the spring **126** of FIGS. **10** and **11**. The spring **226** is analogous to the spring **126**.) By pressing the lever **122** or **222** the toward the cradle **112** or **212**, the lever **122** or **222** assumes a spring force of the unloading-mechanism spring **126** or **226** and increases a length of the receptacle **124** or **224**, which allows the hub **132** or **232** of the needle-in-catheter assembly **130** or **230** to be removed from the receptacle **124** or **224**.

(29) The plunger **116** or **216** of the needle thruster **110** or **210** is configured to move the carriage **114** or **214** within the cradle **112** or **212**. The plunger **116** of the needle thruster **110** includes a longitudinal channel **120** extending along an outer surface of the plunger **116**. The longitudinal channel **120** is configured to receive the longitudinal rail **118** of the cradle **112**. Differently, the plunger **216** of the needle thruster **210** includes a spiral channel **220** spiraling around an outer surface of the plunger **216**. The spiral channel **220** is configured to receive the spiral rail **218** of the cradle **212**.

(30) The plunger **116** or **216** of the needle thruster **110** or **210** is also configured to allow a user to

set the carriage **114** or **214** within the cradle **112** or **212** in accordance with the predetermined length of the needle tract before establishing the needle tract. The needle thruster **210** is different from the needle thruster **110** in that the needle thruster **210** is also configured to allow the user to lock the carriage **214** within the cradle **212** once set in accordance with the predetermined length of the needle tract before establishing the needle tract. As best shown in FIG. **6**, the needle thruster **210** includes a compression spring-loaded lever **228** along a longitudinal side of the cradle **212**. The lever **228** includes a tooth configured to extend through the longitudinal side of the cradle **212** and engage a toothed rack incorporated into the carriage **214** when the spring between the lever **228** and the longitudinal side of the cradle **212** is in its most relaxed state, thereby locking the carriage **214** within the cradle **212**. When the lever **228** is pressed toward the longitudinal side of the cradle **212**, the tooth disengages the toothed rack of the carriage **214** so the plunger can freely move the carriage **214** to set the carriage **214** in accordance with the predetermined length of the needle tract before establishing the needle tract.

(31) With the exception of the compression spring **126** or **226**, which can be made of, for example, stainless steel, the needle thruster **110** or **210** can be made by molding pieces of the needle thruster **110** or **210** and coupling the resulting molded pieces together to make the needle thruster **110** or **210**. The molded pieces can be molded by, for example, injection molding with polyethylene, polycarbonate, or some other medically acceptable thermoplastic. The molded pieces can be coupled together by pressing the molded pieces together if configured with snap-together connectors, adhering the molded pieces together with an adhesive, bonding the molded pieces together with solvent, or a combination thereof.

(32) Adverting to the needle-in-catheter assembly **130** or **230**, the needle-in-catheter assembly **130** or **230** includes the hub **132** or **232**, the catheter tube **134** or **234** extending from a distal-end portion of the hub **132** or **232**, and the needle **137** or **237** disposed within the catheter tube **134** or **234**. (See FIG. **8**.) Each of the hub **132** or **232**, the catheter tube **134** or **234**, and the needle **137** or **237** will now be described.

(33) The hub **132** or **232** of the needle-in-catheter assembly **130** or **230** includes a longitudinal handle **133** or **233** extending from the hub **132** or **232** between a proximal-end portion and the distal-end portion of the hub **132** or **232**. The handle **133** or **233** is configured for handling the needle-in-catheter assembly **130** or **230**. For example, the handle **133** or **233** can be used to manipulate the needle-in-catheter assembly **130** or **230** when using the needle-in-catheter assembly **130** or **230** separately from the needle thruster **110** or **210**, loading the needle-in-catheter assembly **130** or **230** in the needle thruster **110** or **210**, or when unloading the needle-in-catheter assembly **130** or **230** from the needle thruster **110** or **210**.

(34) The hub **132** or **232** of the needle-in-catheter assembly **130** or **230** includes a catheter tube-advancing mechanism for advancing the catheter tube **134** or **234** with respect to the needle **137** or **237**. The catheter tube-advancing mechanism includes a threaded distal-end plug **139** or **239** (not shown) of the hub **132** or **232** of the needle-in-catheter assembly **130** or **230**. (For the distal-end plug **239**, see the distal-end plug **139** of FIG. **12**. The distal-end plug **239** is analogous to the distal-end plug **139**.) The distal-end plug **139** or **239** of the catheter tube-advancing mechanism is configured to advance from the distal-end portion of the hub **132** or **232** upon turning the distal-end plug **139** or **239** in a first direction. Upon turning the distal-end plug **139** or **239** in a second direction, the catheter tube-advancing mechanism is configured to withdraw the distal-end plug **139** or **239** into the distal-end portion of the hub **132** or **232**. Because the catheter tube **134** or **234** is fixedly attached to the distal-end plug **139** or **239**, the catheter tube **134** or **234** is also advanced from the distal-end portion of the hub **132** or **232** upon turning the distal-end plug **139** or **239** in the first direction and withdrawn into the distal-end portion of the hub **132** or **232** upon turning the distal-end plug **139** or **239** in the second direction. The catheter tube-advancing mechanism allows the catheter tube **134** or **234** to be advanced up to or beyond a distal tip of the needle **137** or **237** such as after establishing the needle tract of the predetermined length with the needle-in-catheter

assembly **130** or **230**. The needle **137** or **237** can be subsequently removed from the needle-in-catheter assembly **130** or **230** by disconnecting a compression spring-loaded proximal-end cap **135** or **235** of the hub **132** or **232** of the needle-in-catheter assembly **130** or **230** and withdrawing the needle **137** or **237** from the needle-in-catheter assembly **130** or **230**.

(35) The hub **132** or **232** of the needle-in-catheter assembly **130** or **230** includes a loading mechanism for loading the needle-in-catheter assembly **130** or **230** in the carriage **114** or **214**. The loading mechanism includes the compression spring-loaded proximal-end cap **135** or **235** of the hub **132** or **232** of the needle-in-catheter assembly **130** or **230**. The proximal-end cap **135** or **235** is configured to move toward the distal-end portion of the hub **132** or **232** and compress the spring **136** or **236** (not shown) of the loading mechanism upon loading the hub **132** or **232** in the receptacle **124** or **224** of the carriage **114** or **214**. (For the spring **236**, see the spring **136** of FIGS. **10** and **11**. The spring **236** is analogous to the spring **136**.) A spring force exerted by the loading-mechanism spring **136** or **236** holds the hub **132** or **232** in the receptacle **124** or **224** of the carriage **114** or **214**.

(36) The hub **132** or **232** of the needle-in-catheter assembly **130** or **230** optionally includes a needle-advancing mechanism for advancing the needle **137** or **237** with respect to the catheter tube **134** or **234** or both the needle **137** or **237** and the catheter tube **134** or **234** together with respect the hub **132** or **232**. Such a needle-advancing mechanism can include a threaded needle-advancing mechanism configured to advance the needle **137** or **237** or both the needle and the catheter tube **134** or **234** together upon turning a threaded element (e.g., the proximal-end cap **135** or **235**) of the threaded needle-advancing mechanism in a first direction. Upon turning the threaded element in a second direction, the threaded needle-advancing mechanism is configured to retract the needle **137** or **237** or both the needle and the catheter tube **134** or **234** together. Such a needle-advancing mechanism can alternatively include a slide-based needle-advancing mechanism configured to advance the needle **137** or **237** or both the needle and the catheter tube **134** or **234** together upon sliding a slideable element (e.g., a tab disposed in a longitudinal slot of the handle **133** or **233**) of the slideable needle-advancing mechanism in a first direction. Upon sliding the slideable element in a second direction, the slideable needle-advancing mechanism is configured to retract the needle **137** or **237** or both the needle and the catheter tube **134** or **234** together.

(37) The hub **132** or **232** of the needle-in-catheter assembly **130** or **230** includes an internal chamber **138** or **238** (not shown) in a proximal-end portion of the hub **132** or **232**. (For the internal chamber **238**, see the internal chamber **138** of FIGS. **10** and **11**. The internal chamber **238** is analogous to the internal chamber **138**.) A port **140** or **240** of the hub **132** or **232** and one or more microlumens **141** or **241** longitudinally extending through the catheter tube **134** or **234** are fluidly connected to the internal chamber **138** or **238**. The needle **137** or **237** of the needle-in-catheter assembly **130** or **230** extends through the internal chamber **138** or **238** of the hub **132** or **232** without a fluid connection with the internal chamber **138** or **238** of the hub **132** or **232**. The needle **137** or **237** is fixed to the proximal-end cap **135** or **235** of the hub **132** or **232**.

(38) FIGS. **8** and **9** illustrate different views of a distal-end portion of the needle **137** or **237** disposed in the catheter tube **134** or **234**, in accordance with some embodiments.

(39) The catheter tube **134** or **234** of the needle-in-catheter assembly **130** or **230** is a double-walled catheter tube including an outer wall **142** or **242** and an inner wall **144** or **244**. A distal-end portion of the catheter tube **134** or **234** including a distal end of the catheter tube **134** or **234** has a tapered distal tip including one or more openings corresponding to one or more microlumens **141** or **241** of the catheter tube **134** or **234**. Each of the one or more microlumens **141** or **241** is defined by an inner surface of the outer wall **142** or **242** of the catheter tube **134** or **234**, an outer surface of the inner wall **144** or **244** of the catheter tube **134** or **234**, and one or more struts **146** or **246** longitudinally extending through the catheter tube **134** or **234** between the outer wall **142** or **242** and the inner wall **144** or **244** of the catheter tube **134** or **234**. The one or more microlumens **141** or **241** are configured to allow a bodily fluid such as blood to flash back into the internal chamber **138**

or **238** for drawing the bodily fluid from the port **140** or **240** without withdrawing the needle **137** or **237**.
(40) The catheter tube **134** or **234** of the needle-in-catheter assembly **130** or **230** can include a number of evenly spaced radiopaque rings **148** or **248** on the catheter tube **134** or **234** for radiographically confirming a length of the needle tract after forming the needle tract of the predetermined length by counting of the needle-in-catheter assembly **130** or **230**.
(41) The needle **137** or **237** of the needle-in-catheter assembly **130** or **230** can be a hollow needle or a solid needle as shown. A distal end of the needle **137** or **237** distally extends beyond the distal end of the catheter tube **134** or **234** of the of the needle-in-catheter assembly **130** or **230**. The needle **137** or **237** is disposed within a needle lumen longitudinally extending through the catheter tube **134** or **234**. The needle lumen is defined by an inner surface of the inner wall **144** or **244** of the catheter tube **134** or **234**.
(42) With the exception of the cannula tube **134** or **234**, the needle **137** or **237**, and the compression spring **136** or **236**, which can respectively be made of polyurethane, stainless steel, and stainless steel, the needle-in-catheter assembly **130** or **230** such as the hub **132** or **232** can be made by molding pieces of the needle-in-catheter assembly **130** or **230** and coupling the resulting molded pieces together to make the needle-in-catheter assembly **130** or **230**. The molded pieces can be molded by, for example, injection molding with polyethylene, polycarbonate, or some other medically acceptable thermoplastic. The molded pieces can be coupled together by pressing the molded pieces together if configured with snap-together connectors, adhering the molded pieces together with an adhesive, bonding the molded pieces together with solvent, or a combination thereof.
(43) Methods
(44) Each needle-tract assistant of the needle-tract assistants **100** and **200** is configured for establishing a needle tract of a predetermined length during at least a procedure involving placing a portosystemic shunt between the hepatic vein and the portal vein, which reduces the risk of overshooting the portal vein. As such, the needle-tract assistants **100** and **200** can reduce complications, shorten procedure times, increase success rates, and the like for procedures such as the foregoing procedure.
(45) A method of establishing a needle tract of a predetermined length with the needle-tract assistant **100** or **200** includes inserting the needle-in-catheter assembly **130** or **230** into a stiffening cannula disposed in an introducer sheath positioned in a hepatic vein, the needle-in-catheter **130** or **230** assembly including the needle **137** or **237** having the distal end extending beyond that of the catheter tube **134** or **234** surrounding the needle **137** or **237**; moving the plunger **116** or **216** of the needle thruster **110** or **210** to set the carriage **114** or **214** coupled to the plunger **116** or **216** in position within the cradle **112** or **212** of the needle thruster **110** or **210** for establishing the needle tract of the predetermined length; loading the hub **132** or **232** of the needle-in-catheter assembly **130** or **230** in the carriage **114** or **214** of the needle thruster **110** or **210**; and thrusting the plunger **116** or **216** of the needle thruster **110** or **210** toward a distal end of the needle thruster **110** or **210** to thrust the needle **137** or **237** and catheter tube **134** or **234** of the needle-in-catheter assembly **130** or **230** through a liver parenchyma into a portal vein, thereby forming the needle tract of the predetermined length.
(46) The method further includes advancing the catheter tube **134** or **234** up to or beyond the distal tip of the needle **137** or **237** after establishing the needle tract of the predetermined length by turning the distal-end plug **139** or **239** in the first direction; and removing the needle **137** or **237** from the needle-in-catheter assembly **130** or **230** by disconnecting the proximal-end cap **135** or **235** of the hub **132** or **232** of the needle-in-catheter assembly **130** or **230** and withdrawing the needle **137** or **237** from the needle-in-catheter assembly **130** or **230**.
(47) The method further includes radiographically determining a length for the needle tract of the predetermined length before setting the carriage **114** or **214** of the needle thruster **110** or **210** in

position within the cradle **114** or **214** for establishing the needle tract of the predetermined length.
(48) The method further includes radiographically confirming the length of the needle tract after forming the needle tract of the predetermined length by counting a number of the evenly spaced radiopaque rings **148** or **248** on the catheter tube **134** or **234** of the needle-in-catheter assembly **130** or **230**.
(49) While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

Claims

1. A needle-tract assistant for establishing a needle tract of a predetermined length, comprising: a needle thruster including: a cradle; a carriage within the cradle, the carriage configured to move between a proximal-end portion and a distal-end portion of the cradle; and a plunger coupled to the carriage, the plunger configured to move the carriage within the cradle and allow a user to set the predetermined length of the needle tract before establishing the needle tract; and a needle-in-catheter assembly including: a hub; a catheter tube extending from a distal-end portion of the hub; and a needle disposed within the catheter tube, the needle-in-catheter assembly removably loaded in the carriage of the needle thruster, wherein: a distal end of the needle extends distally beyond a distal end of the catheter tube, and the distal-end portion of the catheter tube includes a tapered distal tip including one or more openings corresponding to one or more microlumens of the catheter tube.
2. The needle-tract assistant of claim 1, wherein the cradle of the needle thruster includes a longitudinal opening between the proximal-end portion and the distal-end portion of the cradle configured to allow the needle-in-catheter assembly to be removed from the needle thruster through the longitudinal opening.
3. The needle-tract assistant of claim 1, wherein the cradle of the needle thruster includes a proximal-end opening at a proximal end of the cradle through which the plunger extends and a distal-end opening at a distal end of the cradle through which the catheter tube and the needle extend when the needle-in-catheter assembly is disposed in the needle thruster.
4. The needle-tract assistant of claim 1, wherein the cradle of the needle thruster includes a longitudinal rail extending along an inner surface of the cradle configured to guide movement of the plunger, the carriage, or both the plunger and the carriage within the cradle.
5. The needle-tract assistant of claim 4, wherein the plunger of the needle thruster includes a longitudinal channel extending along an outer surface of the plunger configured to receive the longitudinal rail of the cradle.
6. The needle-tract assistant of claim 1, wherein the cradle of the needle thruster includes a spiral rail spiraling around an inner surface of the cradle configured to guide movement of the plunger, the carriage, or both the plunger and the carriage within the cradle and govern a linear velocity thereof.
7. The needle-tract assistant of claim 6, wherein the plunger of the needle thruster includes a spiral channel spiraling around an outer surface of the plunger configured to receive the spiral rail of the cradle.
8. The needle-tract assistant of claim 1, wherein the carriage of the needle thruster includes a first component configured to unload the needle-in-catheter assembly from the carriage.
9. The needle-tract assistant of claim 8, wherein the first component includes a compression spring-

loaded lever shaped to form a proximal-end portion of a receptacle in the carriage configured to receive the hub of the needle-in-catheter assembly, the compression spring-loaded lever configured to compress a spring of the first component upon pressing the compression spring-loaded lever toward the cradle, thereby assuming a spring force of an unloading-mechanism spring, increasing a length of the receptacle, and allowing the hub of the needle-in-catheter assembly to be removed from the receptacle.
10. The needle-tract assistant of claim 9, wherein the hub of the needle-in-catheter assembly includes a second component configured to load the needle-in-catheter assembly in the carriage.
11. The needle-tract assistant of claim 10, wherein the second component includes a compression spring-loaded proximal-end cap of the hub of the needle-in-catheter assembly, the compression spring-loaded proximal-end cap configured to move toward the distal-end portion of the hub and compress a spring of the second component upon loading the hub in the receptacle of the carriage, thereby allowing a spring force exerted by the spring of the second component to hold the hub in the receptacle of the carriage.
12. The needle-tract assistant of claim 1, wherein the hub of the needle-in-catheter assembly includes an internal chamber in a proximal-end portion of the hub to which a port of the hub and one or more microlumens longitudinally extending through the catheter tube are fluidly connected.
13. The needle-tract assistant of claim 12, wherein the needle of the needle-in-catheter assembly extends through the internal chamber of the hub without a fluid connection with the internal chamber.
14. A needle-in-catheter assembly for establishing a needle tract, comprising: a hub; a double-walled catheter tube extending from a distal-end portion of the hub, the double-walled catheter tube including an outer wall, an inner wall, and one or more microlumens longitudinally extending through the double-walled catheter tube, each of the one or more microlumens defined by an inner surface of the outer wall of the double-walled catheter tube, an outer surface of the inner wall of the double-walled catheter tube, and one or more struts longitudinally extending through the double-walled catheter tube between the outer wall and the inner wall of the double-walled catheter tube; and a needle disposed within the double-walled catheter tube, the needle fixed to a compression spring-loaded proximal-end cap of the hub.
15. The needle-in-catheter assembly of claim 14, wherein the needle is disposed within a needle lumen longitudinally extending through the double-walled catheter tube, the needle lumen defined by an inner surface of the inner wall of the double-walled catheter tube.
16. The needle-in-catheter assembly of claim 14, wherein the hub includes an internal chamber in a proximal-end portion of the hub to which a port of the hub and the one or more microlumens are fluidly connected.
17. The needle-in-catheter assembly of claim 16, wherein the needle extends through the internal chamber of the hub without a fluid connection with the internal chamber.
18. The needle-in-catheter assembly of claim 14, wherein a distal end of the needle distally extends beyond a distal end of the double-walled catheter tube, a distal-end portion of the double-walled catheter tube including the distal end of the double-walled catheter tube having a tapered distal tip including one or more openings corresponding to the one or more microlumens of the double-walled catheter tube.
19. The needle-in-catheter assembly of claim 14, wherein the compression spring-loaded proximal-end cap of the hub is configured to move toward the distal-end portion of the hub and compress a spring upon loading the hub in a receptacle of a carriage, thereby allowing a spring force exerted by the spring to hold the hub in the receptacle of the carriage.
20. The needle-in-catheter assembly of claim 14, wherein the hub includes a longitudinal handle extending from the hub between a proximal-end portion and the distal-end portion of the hub.
21. A needle-tract assistant for establishing a needle tract of a predetermined length, comprising: a needle thruster including: a cradle; a carriage within the cradle, the carriage configured to move

between a proximal-end portion and a distal-end portion of the cradle, the carriage including a first component configured to unload a needle-in-catheter assembly from the carriage; and a plunger coupled to the carriage, the plunger configured to move the carriage within the cradle and allow a user to set the predetermined length of the needle tract before establishing the needle tract; and the needle-in-catheter assembly including: a hub including a second component configured to load the needle-in-catheter assembly in the carriage; a catheter tube extending from a distal-end portion of the hub; and a needle disposed within the catheter tube, the needle-in-catheter assembly removably loaded in the carriage of the needle thruster, wherein: the first component includes a compression spring-loaded lever shaped to form a proximal-end portion of a receptacle in the carriage configured to receive the hub, and the compression spring-loaded lever is configured to compress a spring of the first component upon pressing the compression spring-loaded lever toward the cradle, thereby assuming a spring force of an unloading-mechanism spring, increasing a length of the receptacle, and allowing the hub of the needle-in-catheter assembly to be removed from the receptacle.

22. A needle-tract assistant for establishing a needle tract of a predetermined length, comprising: a needle thruster including: a cradle; a carriage within the cradle, the carriage configured to move between a proximal-end portion and a distal-end portion of the cradle; and a plunger coupled to the carriage, the plunger configured to move the carriage within the cradle and allow a user to set the predetermined length of the needle tract before establishing the needle tract; and a needle-in-catheter assembly including: a hub including an internal chamber in a proximal-end portion of the hub; a catheter tube extending from a distal-end portion of the hub, wherein a port of the hub and one or more microlumens longitudinally extending through the catheter tube are fluidly connected to the internal chamber; and a needle disposed within the catheter tube, the needle-in-catheter assembly removably loaded in the carriage of the needle thruster.

23. The needle-tract assistant of claim 22, wherein the needle of the needle-in-catheter assembly extends through the internal chamber of the hub without a fluid connection with the internal chamber.